Six boat counts

What holds a paper boat?

Name _____ Date _____

All numbers are pretend. Show your tens and ones or write your steps.

1. Little gathers 28 pennies, then 17 more beside the tub. How many pennies are ready?

28 + 17 = _____

Show your thinking.

2. Little has 52 pennies beside the tub and puts 26 away. How many remain?

52 − 26 = _____

Show your thinking.

3. Little counts 36 folds across several boats, then 27 more folds. How many folds in all?

36 + 27 = _____

Show your thinking.

Six boat counts

What holds a paper boat?

Name _____ Date _____

All numbers are pretend. Show your tens and ones or write your steps.

4. Little draws 71 drops beside a boat sketch, then erases 38. How many drawn drops remain?

71 − 38 = _____

Show your thinking.

5. Little draws 24 waterline marks, then follows “Add 19 marks.” How many marks are drawn?

24 + 19 = _____

Show your thinking.

6. A pretend boat voyage lasts 63 minutes. After 28 minutes, how many minutes remain?

63 − 28 = _____

Show your thinking.

WONDER DAILY

What holds a paper boat?

What holds a paper boat?



One leaf. Six steps. A story that keeps going.

GRADE 3

Little Wonderer set a sheet of paper beside a tub. "Could this carry a penny?" Big Wonderer put down a short list of folding commands. Little traced the first line. The paper was flat and flimsy. Somehow, those steps would have to give it a new shape.



"Fold the paper in half. Fold the corners down." Little followed each command in order. More folds made an open boat with walls and a broad bottom. "Like a program," Little said, "each step helps the next one work." Now came the test. Would the folds hold?



Little lowered the boat into the tub. It floated: it stayed at the water's surface. The boat pushed water aside, or displaced it. Water pushed back up. That push equaled the weight of the displaced water. When the upward push balanced the boat's weight, the boat stopped sinking lower.



Little added a penny, then another. This load was the weight the boat carried. The boat sank lower, displacing more water. Its waterline, where water met the paper, climbed up the sides. Folded walls kept water out. The broad bottom and air-filled space inside helped the boat displace enough water.





"Add one penny. Watch the waterline," Little wrote. Then one more penny made the rim dip under. Water spilled inside. The boat was swamped! A flat sheet had no walls to protect an air-filled space. Shape mattered, and even this folded boat could not carry an endless load.



Later, the boat sagged. Paper's tiny plant fibers absorb water: they take it in. Wet paper grows heavy and soft until its folds lose their shape. A crayon or wax coat slows soaking. Little saved the commands. "Could my next boat last longer?"

Make a little book.

1. Print one-sided: US Letter, portrait, actual size / 100%. No fit. Headers/footers off.
2. Fold long edge to long edge; reopen. Fold short edge to short edge, then in half again. Open: eight boxes.
3. Cut only the solid center line across the middle two boxes. Keep the outer boxes joined.
4. Refold long edge to long edge, print outside. Push ends together to open a diamond, then a cross.
5. Wrap into a book with page 1 in front. Follow pages 1 to 8.



Pictures are simplified and not to scale. Look closely, wonder, and pause whenever you like.

Your turn

Write one command sentence for Little's boat. Tell what to do.

Example: First fold the paper in half, then fold the corners down.

Give the boat an instruction.

What holds a paper boat? · Curiosity day

Name _____ Date _____

Choose one scene. Write one clear command sentence for Little's boat. You may add a second step.



The folding commands



A climbing line



A soft ending

Draw the boat before or after your command. Add one label.

fold · boat · tub · penny · waterline · load · absorb · first · then

First, ... tell how to begin folding

Then, ... tell what to add or watch

Next time, ... tell what to change or compare

Read your sentence. Does it tell what to do, using a command such as Fold, Add, Watch, or Compare?